

Operasi Hitung Campuran (BODMAS/PEMDAS) - Kheylya

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Student

Student: Kheylya

Topic: Operasi Hitung Campuran (BODMAS/PEMDAS)

Status: completed

Lesson Plan

Learning Objectives

- Calculate numerical expressions by applying the correct order of operations (BODMAS).
- Solve mathematical expressions involving brackets, multiplication, and division in the correct sequence.
- Construct mathematical expressions based on simple real-world word problems.
- Compare the results of expressions when brackets are placed in different positions.

Lesson Information

Lesson Sections

1. Spiral Review Warm-up (10 minutes)

Teacher Script:

Hi Kheylya! Welcome back to our math lab. Before we start our new mission, we need to calibrate our detective tools. Let's do a quick 'Number Recall'. For our first task: Look at the number 4,521. What is the value of the digit 5? Type it in the chat! Great, now for the second one: If I have 3 blocks of a thousand and 7 blocks of ten, what is the total value? Next, a quick brain teaser: If we play our number game and I say 'Double 15 and subtract 5', what's my secret number? Last one: What is the largest 4-digit number you can make using 2, 8, 0, and 5?

Activities:

- *Chatbox speed round: Place Value identification.*
- *Visual block value calculation.*
- *Mental math 'Number Game' challenge.*

2. Introduction & Real-World Connection (5 minutes)

Teacher Script:

Imagine you are a detective opening a high-security vault. If you enter the code numbers in the wrong order, the alarm goes off! Math is exactly like that. If we solve a big equation in the wrong order, we get the wrong answer. Today, we are learning the 'Secret Code' called BODMAS. It tells us exactly which part of the math problem to solve first so we can unlock the right answer every single time.

Activities:

- *Storytelling about high-security codes.*
- *Visual reveal of the BODMAS acronym.*

3. Review of Prior Knowledge (5 minutes)

Teacher Script:

Before we dive in, let's look at this simple one: $5 + 2 \times 3$. If I just go left to right, I get 21. But if I do the multiplication first, I get 11. Which one feels 'more' correct in math land? In our previous diagnostic, you were great at multiplication. Today, we're going to use those skills as our priority tool!

Activities:

- *Comparing two different results for the same problem.*
- *Defining 'Operation' (Plus, Minus, Multiply, Divide).*

4. Problem Introduction (10 minutes)

Teacher Script:

Let's meet BODMAS. B stands for Brackets (), O is for Orders (like squares, but we focus on brackets today), D and M are Division and Multiplication, and A and S are Addition and Subtraction. Think of it as a ladder. We start at the top with Brackets. Let's look at this: $(10 + 2) \times 5$. Based on BODMAS, the brackets are the bosses. We must do $10 + 2$ first! What do we get? Yes, 12! Then 12×5 is 60. See how the brackets changed the game?

Activities:

- *Interactive breakdown of the BODMAS acronym.*
- *Step-by-step walk-through of a bracket expression.*

5. Guided Practice (15 minutes)

Teacher Script:

Now, let's solve some mysteries together. Mission 1: $50 - (10 + 5) \times 2$. I want you to circle the part we do first on the screen. Excellent, the brackets! Now, what is $10 + 5$? Write it underneath. Now we have $50 - 15 \times 2$. Look at our BODMAS ladder... do we subtract or multiply first? Correct, Multiply! 15×2 is 30. Finally, 50 minus 30 is... 20! You're a natural detective, Kheyla.

Activities:

- *Solving 3 expressions of increasing difficulty with teacher guidance.*
- *Annotation on the whiteboard to circle the 'first step'.*

6. Independent Practice (15 minutes)

Teacher Script:

It's time for the 'Solo Detective Challenge'. I have placed 5 locked briefcases on the board. Each one has a math expression. Solve them one by one. If you get stuck, call for backup! Try this one first: $24 / (6 - 2) + 7$.

Activities:

- *Solving 5 problems independently.*
- *A small word problem: 'Kheyla bought 3 boxes of 10 crayons, then gave 5 to a friend. Write the math sentence.'*

7. Consolidation & Reflection (5 minutes)

Teacher Script:

Amazing work today, Detective Kheyla! You've mastered the order of operations. Remember: Brackets always come first. Before you go, tell me: what was the trickiest part today? One more thing—make sure to join <https://learn.algonova.id/login> to see your rewards and finish today's mission. See you next time!

Activities:

- *Reflective questioning.*
- *Login reminder.*

Homework

Medium Questions:

1. $12 + (15 / 3)$
2. $(20 - 5) \times 4$
3. $40 / (2 + 3) \times 2$
4. $100 - (25 \times 3) + 10$

Hard Questions:

1. $50 + (10 \times 2) / (12 - 7)$
2. $((15 + 5) \times 2) - (10 / 2)$
3. Write the expression and solve: Multiply 5 by the sum of 10 and 2.
4. $8 \times 4 / (10 - 6) + 15$
5. Compare: Is $(10 + 5) \times 2$ the same as $10 + (5 \times 2)$? Explain why.
6. Challenge: $144 / (12 \times 1) + (5 - 2) \times 10$

Answer Key:

- `1. 17 (15/3=5, 12+5=17)`

`2. 60 (20-5=15, 15\*4=60)`
 `3. 16 (2+3=5, 40/5=8, 8\*2=16)`
 `4. 35 (25\*3=75, 100-75=25, 25+10=35)`
 `5. 54 (10\*2=20, 12-7=5, 20/5=4, 50+4=54)`
 `6. 35 (20\*2=40, 10/2=5, 40-5=35)`
 `7. 5 x (10 + 2) = 60`
 `8. 23 (10-6=4, 8\*4/4=8, 8+15=23)`
 `9. No. (10+5)\*2=30 vs 10+(5\*2)=20. Brackets change the sum order.`
 `10. 42 (12\*1=12, 144/12=12, 5-2=3, 3\*10=30, 12+30=42)`
 `REVIEW: Nilai Tempat - 5,902. Value of 9? Answer: 900 (Hundreds).`
 `REVIEW: Nilai Tempat - Write 7,000 + 400 + 5 in standard form. Answer: 7,405.`
 `REVIEW: Nilai Tempat - Compare 3,450 and 3,540. Answer: 3,540 is greater.`
 `REVIEW: Nilai Tempat - What is 1,000 more than 8,230? Answer: 9,230.`
 `REVIEW: Diagnostik - If you double a number and get 40, what is the number? Answer: 20.`
 `REVIEW: Diagnostik - Which is smaller: 1/2 of 100 or 3 x 15? Answer: 3 x 15 (45) is smaller than 50.`
 `REVIEW: Diagnostik - I have 4 legs in the morning and 2 in the afternoon. If I have 10 people, how many legs in the morning? Answer: 40.`
 `REVIEW: Diagnostik - Missing number: 2, 5, 8, \_\_, 14. Answer: 11 (+3 pattern).`

Parent Reminder

Today's Math Mission: The Secret Code of BODMAS

Kheyla did a great job today acting as a math detective! We learned the order of operations (BODMAS) to solve complex problems. She practiced identifying which parts of a calculation to solve first—especially when brackets are involved. Your detective has homework waiting to sharpen these skills!